

An Open Source Tool for Lifetime Estimation of IGBT-Diode Power Modules

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Introduction: Why Converter Reliability Matters

- ▶ Inverters are now the backbone of PV, wind and battery systems.
- ▶ But they are one of the weakest link:
 - ▶ 43–70% of PV plant service requests.
 - ▶ 0.48 converter failures per wind turbine year.
- ▶ Modern inverters must also provide grid support (reactive power, voltage, fault ride through).
- ▶ This adds additional electrical + thermal stress.

Introduction: The IGBT-Diode Module

- ▶ IGBT-diode power modules form the switching core of modern converters.
- ▶ Dominant failure mechanism: thermo-mechanical fatigue
 - ▶ Driven by repetitive junction temperature cycling.
 - ▶ > 55% of power electronics failures are temperature induced.
- ▶ Cycling comes from fluctuating electrical load + environment.
- ▶ Progressively degrades bond wires, solder layers, interconnects.

Introduction: Identified Research Gap

- ▶ No unified, open, reproducible framework maps real mission profiles → component lifetime.
- ▶ Existing approaches are fragmented:
 - ▶ FEM / aging tests (ANSYS, COMSOL): accurate but too slow for long profiles.
 - ▶ Manufacturer tools (IPOSIM, SemiSel): closed source, device locked.
 - ▶ ML / RUL libraries (CeRULEo, ProgPy): no physics, device to mission chain.

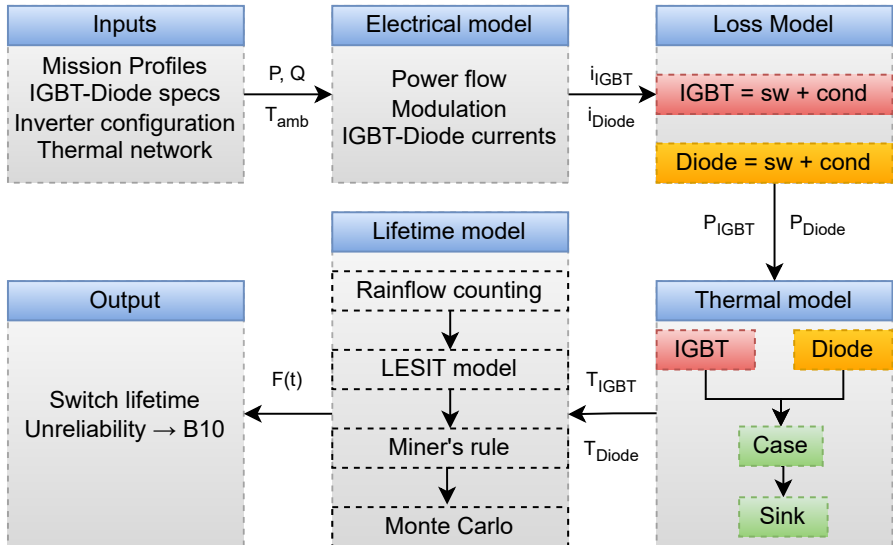
The Gap

An accessible, mission profile driven reliability framework → PEARL

Introduction: Research Contribution

- ▶ PEARL: an open source electro-thermal reliability framework for power electronics.
 - ▶ This work focuses on IGBT-Diode power modules.
- ▶ A unified, reproducible workflow linking:
 - ▶ Mission profiles, operating conditions, design + component specs.
 - ▶ → semiconductor lifetime estimation.
- ▶ Integrates different domains in one pipeline:
 - ▶ Electrical modelling
 - ▶ Loss modelling
 - ▶ Thermal modelling
 - ▶ Lifetime modelling
- ▶ Transparent, scalable, computationally efficient.

Methodology: Workflow



Methodology: Electrical Modelling

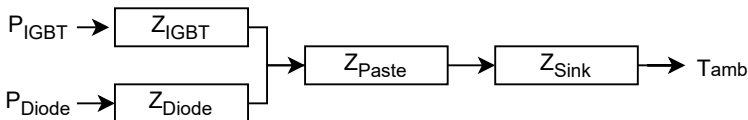
- ▶ Inputs of mission profile operating points + inverter configuration.
- ▶ Power flow: convert P , Q into operating point.
- ▶ Reconstruct instantaneous device currents:
 - ▶ IGBT = positive half, Diode = negative half of leg current
- ▶ Output: time-series current profiles $i_{IGBT}(t)$, $i_{Diode}(t)$.

Methodology: Loss Modelling

- ▶ Losses computed from instantaneous device currents.
- ▶ IGBT:
 - ▶ Switching (turn on + turn off) + conduction losses
- ▶ Diode:
 - ▶ Reverse recovery + conduction losses
- ▶ Total: $P_I = P_{sw,I} + P_{con,I}$, $P_D = P_{sw,D} + P_{con,D}$

Methodology: Thermal Modelling

- ▶ Losses P_I , P_D injected into a multi stage Cauer RC network.
- ▶ Four coupled parts: IGBT ladder, Diode ladder, shared paste + heat sink.
- ▶ IGBT and Diode thermally coupled through the shared case.
- ▶ Solve coupled ODEs $\rightarrow$ junction temperatures $T_{j,I}(t)$, $T_{j,D}(t)$.



Methodology: Lifetime Modelling

- ▶ Rainflow counting $\rightarrow$ thermal cycles $(\Delta T_j, T_{jm}, t_{on}, \text{count})$.
- ▶ Map each cycles to failure N_f (LESIT power cycling model).
- ▶ Accumulate damage via Miner's rule $\rightarrow$ lifetime (years).
- ▶ Monte Carlo on parameter uncertainty $\rightarrow$ unreliability $F(t)$, B10.

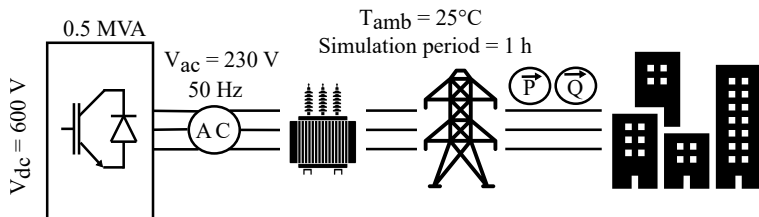
$$N_f = A_0 A_1 e^{-\frac{\Delta T_j - T_0}{\lambda}} (\Delta T)^\alpha e^{-\frac{\Delta T_j - T_0}{\lambda}} \times e^{\frac{E_a}{k_B T_{jm}}} \left(\frac{C + t_{on}^\gamma}{C + 2^\gamma} \right) k_{\text{thickness}}$$

Key drivers: $\downarrow N_f$ with larger swing ΔT_j and higher mean temperature T_{jm}

Case Study 1: Research Question & Setup

Question

How does reactive power affect module stress and lifetime?

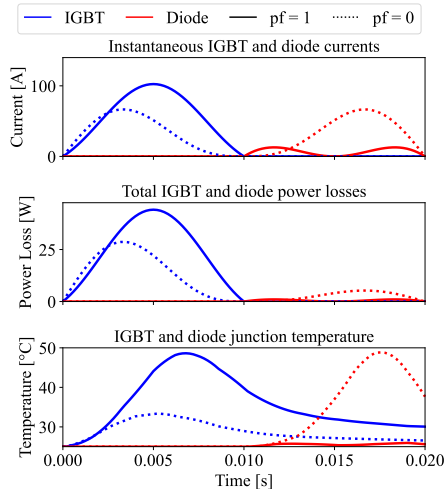


Case Study 1: Setup

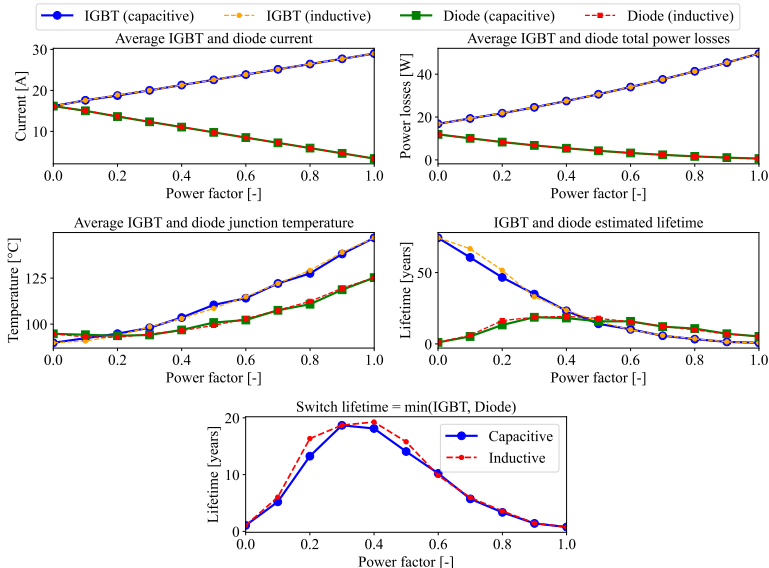
- ▶ Three-phase, 0.5 MVA, 230 V AC / 600 V DC, 10 kHz, 50 Hz.
- ▶ Device: Infineon IKW50N60H3 (600 V, 50 A) $\times 10$ in parallel
- ▶ Paste: Thermal Grizzly Kryonaut
- ▶ Heat sink: Cooling Innovations 3-121211U
- ▶ Sweep power factor 0 to 1 (inductive + capacitive), fixed apparent power.
- ▶ 1 hour profile, $T_{amb} = 25^{\circ}\text{C}$, repeated over service life.

Case Study 1: One Grid Cycle — Currents to Temperature

- Currents → losses → temperature
- pf= 1: IGBT hot
- pf= 0: diode hotter (higher R_{th})



Case Study 1: Reactive Power Reshapes Stress & Lifetime



Case Study 2: Research Question & Setup

Question

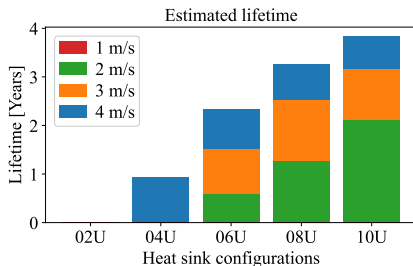
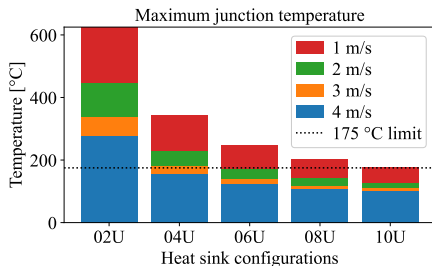
How does heat sink choice affect module's lifetime & reliability?

Setup:

- ▶ Same conditions as Case Study 1, but power factor fixed at $\text{pf} = 1$.
- ▶ Device: Infineon IKW50N60H3 (600 V, 50 A) $\times 10$ in parallel
- ▶ Constraint: Junction temperature $\leq 175^{\circ}\text{C}$ (datasheet limit)
- ▶ Sweep heat sinks: Cooling Innovations 3-1212 series (02U, 04U, 06U, 08U, 10U)
- ▶ Differ in height, weight & thermal resistance vs. air speed.

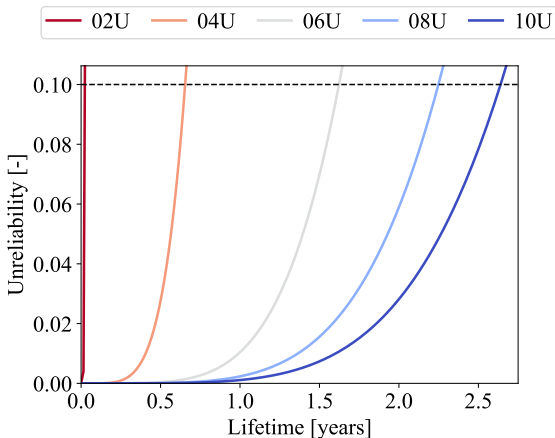
Case Study 2: Results & Design Trade-off

- ▶ Larger sink $\rightarrow$ lower thermal resistance $\rightarrow$ lower temperature.
- ▶ Lower junction temperature $\rightarrow$ longer lifetime: 10U at 4 m/s is coolest and lives longest.
- ▶ Trade-off: compact sinks save space but need stronger forced cooling (larger fans, more auxiliary power).



Case Study 2: Heat Sink Impact on Unreliability

- Unreliability $F(t)$ = probability the switch has failed by time t where dashed line marks B10 (10% failed).



Conclusion

- ▶ PEARL unifies electrical loss, dynamic thermal and power cycling lifetime models into one workflow for IGBT-diode modules.
- ▶ Datasheet driven: translates mission-profile operating conditions into cumulative lifetime consumption.
- ▶ Case Study 1 - Reactive power: At constant apparent power, power factor alone reshapes IGBT vs diode lifetime (peak near $\text{pf} \approx 0.4$).
- ▶ Case Study 2 - Heat sink: Sink size & air flow directly set junction temperature and resulting lifetime, guiding early design choices .
- ▶ Transparent link from mission profiles & design configurations → semiconductor lifetime, enabling reliability aware design

Future Outlook

- ▶ Extend to more components: DC-link capacitors, LCL filters, MOSFETs, etc.
- ▶ Incorporate their lifetime models → system level reliability of grid connected converters.
- ▶ Long term: A flexible platform for diverse power electronic assets (STATCOMs, HVDC converters, etc).
- ▶ Enable grid wide reliability evaluation under varying operating conditions.

Real World Applications: LCL Example

